

1) Data Generation

Random Sampling:
 m, h, t, g, P
(uniform distributions)

Compute Targets:
 $KE = P \cdot (h/t)$
 $PE = m \cdot g \cdot h$

Add Gaussian Noise:
(0.1%–1% of KE, PE)

2) Dataset Construction

$X = [KE_noisy, t, h, g]$ or $[m, h, t, P, g]$
 $Y = [PE_noisy]$ or $[KE_noisy]$
 $Y \rightarrow \text{kg} \cdot \text{m}^2/\text{s}^2$

3a) Symbolic Regression (PySR) - KE

PySRRegressor
Ops: +, -, *, /, ^, sqrt, log, exp, sin
Iterations: 500
Parsimony: 0.0001

4a) KE Formula & Metrics

KE Formula:
 $(P \cdot h)/t$
 $R^2: 0.99993$
MAE: 4.32
MSE: 139.03